

Identifying, Comparing, and Quantifying Entanglement in Two-Qubit Systems

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Introduction

Entanglement is a defining feature of composite quantum systems and a central resource in quantum information theory. This document presents intuitive and structural methods for identifying entanglement in two-qubit states, comparing degrees of entanglement, and understanding how entanglement behaves under noise.

General Two-Qubit State

A general pure state of two qubits is written as

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \quad (1)$$

with complex coefficients satisfying normalization.

Separable vs. Entangled States

A pure state is **separable** if it can be written as

$$|\psi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle. \quad (2)$$

Otherwise, it is **entangled**. This distinction can often be made by inspection.

Schmidt Decomposition (Most Elegant Criterion)

Statement of the Theorem

Any pure state of two qubits can be written as

$$|\psi\rangle = \lambda_1 |u_1\rangle |v_1\rangle + \lambda_2 |u_2\rangle |v_2\rangle, \quad (3)$$

where:

- $\lambda_1, \lambda_2 \geq 0$
- $\lambda_1^2 + \lambda_2^2 = 1$
- $\{|u_i\rangle\}$ and $\{|v_i\rangle\}$ are orthonormal bases

Entanglement Classification

- If $\lambda_1 = 1, \lambda_2 = 0$: **separable**
- If $0 < \lambda_2 < \lambda_1 < 1$: **partially entangled**
- If $\lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}$: **maximally entangled**

Key Insight

A pure two-qubit state is maximally entangled if and only if its Schmidt coefficients are equal.

This immediately explains why Bell states are the most entangled states.

Maximally Entangled States

The Bell states

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle) \quad (4)$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \quad (5)$$

have equal Schmidt coefficients and are therefore maximally entangled.

Each subsystem alone appears completely random; all information resides in correlations.

Mixed States and the PPT Criterion

Peres–Horodecki (PPT) Criterion

For a bipartite density matrix ρ , define the *partial transpose* with respect to subsystem B:

$$\rho^{T_B}.$$

- If ρ^{T_B} has a negative eigenvalue $\Rightarrow$ **entangled**
- If $\rho^{T_B} \geq 0$ (PPT), the state may or may not be entangled

Special Case: Two Qubits

For 2×2 systems:

PPT is both necessary and sufficient for separability.

Thus, for two qubits, positivity of the partial transpose completely determines entanglement.

Physical Intuition

Partial transposition acts like a “time-reversal” on one subsystem. Entangled states fail this test because their correlations are incompatible with any classical probabilistic description.

Werner States

Werner states model entanglement mixed with white noise:

$$\rho_W(p) = p |\Psi^-\rangle \langle \Psi^-| + (1-p) \frac{\mathbb{I}}{4}, \quad (6)$$

where $0 \leq p \leq 1$.

Entanglement Properties

- $p = 1$: maximally entangled
- $p = 0$: completely mixed (separable)
- Entangled if and only if $p > \frac{1}{3}$

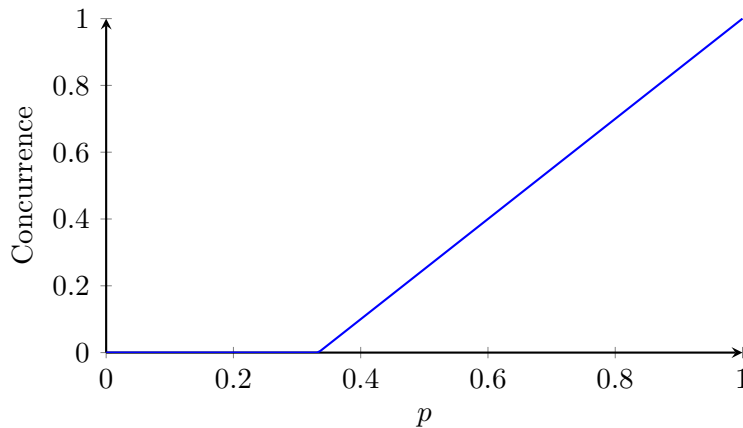
This threshold is detected exactly by the PPT criterion.

Entanglement vs. Werner Parameter

Conceptual Behavior

- Entanglement decreases monotonically with noise
- Vanishes sharply at the critical value $p = \frac{1}{3}$

Example: Concurrence vs. p



The concurrence is zero for $p \leq \frac{1}{3}$ and increases linearly thereafter, reaching its maximum at $p = 1$.

Summary

- Schmidt decomposition provides the cleanest classification of pure-state entanglement
- Bell states are maximally entangled due to equal Schmidt coefficients
- PPT criterion fully characterizes two-qubit mixed-state entanglement

- Werner states demonstrate how entanglement degrades under noise

Central insight: Entanglement is not about superposition alone, but about how information is distributed across subsystems and survives noise.